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Neural systems for processing social relationship information along two principal dimensions

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Comprehending the social relationships of others is essential for daily social interactions. Previous research has revealed the neural substrates involved in inferring others' mental states and detecting social features. However, the neural bases underlying the processing of complex information about diverse human social relationships remain poorly understood. Here, using empirical data collected while participants viewed naturalistic movie clips and evaluated social relationships inside an fMRI scanner, we identify two primary dimensions underlying social relationship recognition. The first dimension reflects Amity–Hostility, while the second dimension encompasses Restrained Amity–Suppressive Hostility. Moreover, the first dimension is predominantly processed within a posterior cortical network centered around the posterior superior temporal sulcus (pSTS), while the second dimension is primarily processed within an anterior cortical network centered around the ventromedial prefrontal cortex (vmPFC). These findings suggest a framework for understanding the neural bases of human social recognition, emphasizing information-based processing along two principal dimensions, each engaging distinct neural systems in the brain.

Understanding the social relationships of other people is essential for our daily social interactions. By observing and recognizing others' social relationships, we can learn social behaviors and determine how we interact with people and social networks. How, then, can we effectively ascertain complex interpersonal relationships in our society? Is there a common dimensional organization for assessing the nature and extent of such relationships in the brain?

Conceptual models, such as the interpersonal circumplex[1–3] have been proposed for classifying and measuring individual interpersonal tendencies. The interpersonal circumplex model conceptualizes social relationships as varying along two primary axes: hostility versus friendliness (affiliation) and dominance versus submissiveness (autonomy). The Structural Analysis of Social Behavior (SASB) model extends this framework, describing social relationships in terms of the affiliation axis, with love and hate going in opposite directions, and autonomy axis, with independence and submission going in opposite directions, across three surfaces that correspond to the subjects of action and reaction[1,4–6]. These models commonly suggest characterizing social relationships based on the conceptually orthogonal dimensions of affiliation and autonomy. Additionally, another stream of social cognition research suggests two universal dimensions, warmth and competence, that are quite similar, although not conceptually identical to these theoretical models[7,8]. However, the question remains whether we do indeed process the information of social relationships commonly based on these two dimensions and whether such dimensional organization exists in the brain.

To understand the neural mechanisms underlying social cognition, prior studies in neuroscience have focused on the processes involved in inferring the mental states and intentions of both others and oneself—a cognitive ability known as theory of mind[9,10]. Experience sharing theory (or simulation theory) suggests that inferring others' thoughts are based on our ability to mimic their actions[11–15]. Neuronal recording and neuroimaging studies have suggested that shared responses between action execution and observation in the mirror neuron network, including the premotor cortex (PMC), the inferior parietal lobule (IPL), medial frontal cortex (MFC), ventrolateral prefrontal cortex (vlPFC), and anterior cingulate gyrus (ACG), provide the neural basis for understanding others' actions and thoughts[16–18]. On the other hand, mental state attribution theory (or theory theory) considers situations in which others' observable cues do not match their mental states[19,20]. The regions including the anterior and posterior part of the superior temporal sulcus (STS), medial prefrontal cortex (mPFC), and

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right temporoparietal junction (rTPJ) have been reported to be engaged in various tasks in which participants are making inferences about others' thought[21–26]. Additionally, posterior regions including the pSTS have been proposed as part of the visual stream involved in perceiving visual stimuli depicting social interactions between individuals[27,28] as well as social features such as facial expressions[29–31]. However, although prior studies have identified the neural substrates involved in inferring others' mental states or detecting social features, a substantial gap remains in our understanding of the neural bases underpinning the processing of social relationship information, such as love or submission, in the brain.

In this study, we investigate dimensional processing in the brain during the recognition of social relationships. While previous literature has conceptually delineated an established organizational framework, wherein affiliation and autonomy are considered two primary orthogonal dimensions[32,33], some empirical evidence also suggests that these dimensions may not be entirely orthogonal and can exert mutual influences[21,34]. Additionally, real-life ambivalent relationships that encompass both love and resentment raise the question of whether love and hate truly lie on opposite ends of a single dimension.

To determine the principal dimensions underlying the recognition of social relationships and to elucidate the neural bases of these dimensions, we designed a functional magnetic resonance imaging (fMRI) task. During the task, participants viewed movie clips depicting various interpersonal relationships and rated each given clip based on four dimensions: love, hate, independence, and submission (Fig. 1a). By measuring these dimensions in parallel, we aimed to empirically assess whether individuals recognize social relationships primarily along the affiliation (love–hate) and autonomy (independence–submission) dimensions, or whether alternative dimensions underlie social relationship recognition. All clips were edited from a single movie and arranged sequentially, allowing the narrative to flow naturally from one trial to the next, thereby simulating the recognition of interpersonal relationships in realistic social contexts. Utilizing principal component analysis (PCA) based on the behavioral ratings, we reveal two primary orthogonal dimensions for the recognition of social relationships, which are consistent across participants. These dimensions of social relationship information are processed in distinct cortical regions and neural networks.

Results

Primary dimensions of social relationship recognition

During the experiment, participants watched the edited version of the movie, *Actress*, and rated each given clip based on the four categories of social relationship, love, hate, independence and submission, on a scale from 0 to 10 (Fig. 1a, b). We first assessed participants' recognition of each category while they were viewing the clips. We compared both the average number of clips that were rated 5 or higher out of 10 (Supplementary Fig. 1a)

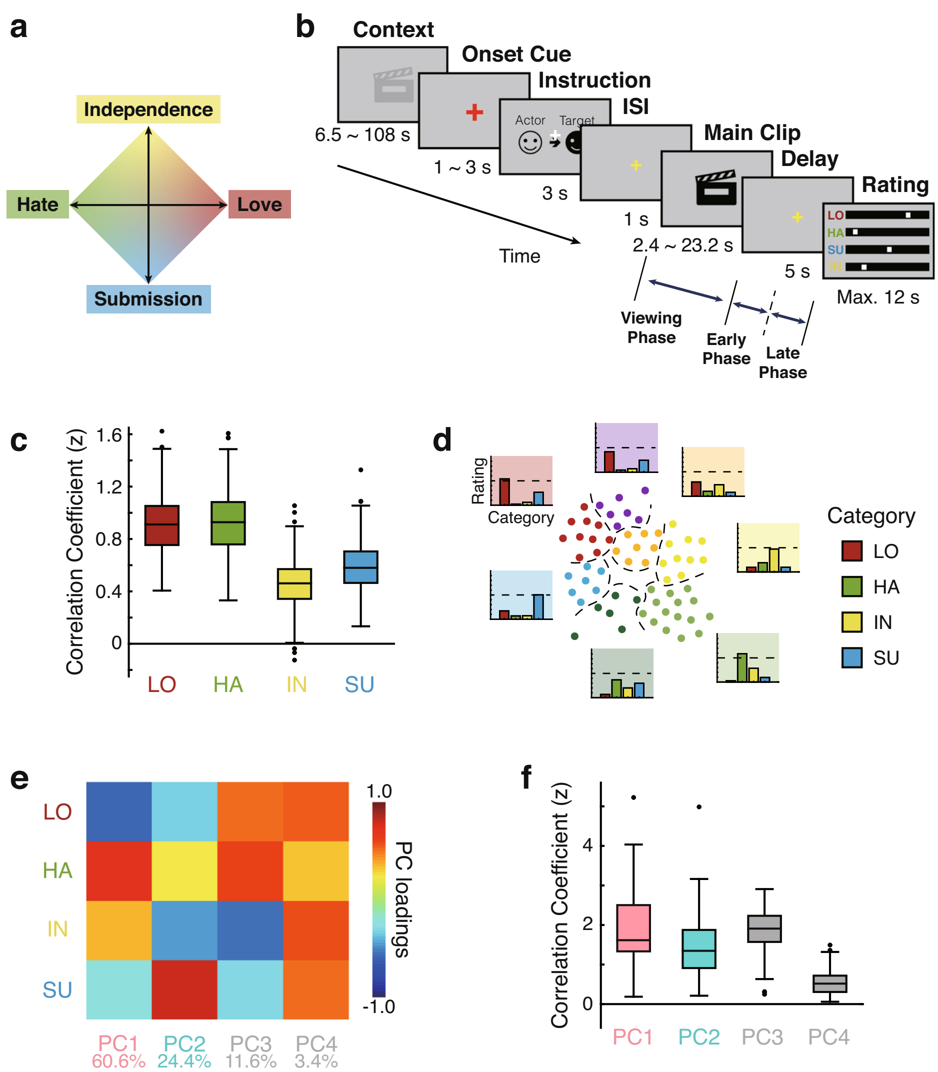

Fig. 1 | Experimental design and principal dimensions of social relationship recognition. **a** Four categories of social relationships–love, hate, independence, and submission–were assessed by participants during the social recognition task. **b** During the social recognition task, participants viewed movie clips and evaluated the social relationships between the actor and the target person, from the actor's perspective. Each trial of the task began with a context clip, providing background information on the social situation. After the context, participants were presented with an onset cue (a red fixation cross with a beep sound), followed by the instruction phase, which informed the identities of the actor and the target. This was followed by an inter-stimulus interval (ISI). Subsequently, the main clip was shown, followed by a delay and then rating phase. In the rating phase, participants rated the main clip using an eleven-point scale for each of the four categories: love (LO), hate (HA), independence (IN), and submission (SU). **c** Boxplots depicting inter-participant agreement for each of the four categories: love (LO), hate (HA), independence (IN), and submission (SU). **d** Clustering analysis results. Each dot represents a specific main clip with dot colors indicating individual clusters. The average ratings for the four categories within each cluster are shown as bar graphs, using the same background color as their corresponding cluster. The dashed line in the bar graphs represents a rating scale of 5. **e** Principal component analysis (PCA) results: The loadings of the four principal components are illustrated. The variance explained by each component is 60.6% for PC1, 24.4% for PC2, 11.6% for PC3, and 3.4% for PC4. **f** Boxplots of agreement between participants for each of the four principal components. In (**c**) and (**f**), the central line represents the median correlation across participants, and the bottom and the top of the box correspond to the 25th and 75th percentiles, respectively. The whiskers extend to the largest and smallest values within 1.5 times the interquartile range from the hinges, with outliers beyond this range plotted as individual dots.

and the average rating scores for each type of social relationship (Supplementary Fig. 1b), and found that both the average number of clips and the average rating level were comparable across the social relationship categories (one-way ANOVA, $F_{(1.795,\ 52.064)} = 2.836$, $p = 0.073$ for the average number of clips; $F_{(3,\ 87)} = 1.401$, $p = 0.248$ for the average ratings). These indicate the participants' comparable levels of recognition for each social relationship category while viewing the movie. We also examined the variability of ratings across participants for each category and found that all four categories of social relationships were consistent across the participants ($t_{(434)} = 91.009$, $p < 0.001$ for love; $t_{(434)} = 83.799$, $p < 0.001$ for hate; $t_{(434)} = 64.362$, $p < 0.001$ for submission; $t_{(434)} = 51.464$, $p < 0.001$ for independence), indicating significant agreement in participants' ratings for social relationships (Fig. 1c).

In real-life scenarios, individuals may encounter situations where a combination of two or more categories of social relationships is recognized, as well as the situations where one of the four basic categories is distinctly recognized. To examine patterns of social recognition, we conducted a clustering analysis (See "Methods" for details) based on the participants' ratings, and identified seven distinct states of social relationship recognition throughout the movie (Fig. 1d), suggesting a diverse and complex range of social relationships beyond the four basic categories.

To directly assess the coexistence of the basic categories, we also analyzed the occurrence of each pair of categories based on participants' ratings of social relationship recognition. Notably, the simultaneous recognition of two social relationship categories was observed across all pairs of the four categories (Supplementary Fig. 2). Even in relationships traditionally regarded as bipolar, such as love–hate or independence–submission, instances of simultaneous recognition were observed. For example, concurrent recognition of love and hate was observed when an actor smiled at a target who consistently refused to cooperate. An example of simultaneous recognition of independence and submission was when an actor put out a cigarette at the target's request, despite displaying facial expressions suggesting reluctance to comply.

To quantitatively assess the relationships between the categories, we conducted regression analyses and derived linear regression lines for the love–hate and independence–submission axes based on participants' subjective ratings. The regression model revealed a negative relationship between love and hate ($\beta = -0.620$, $p < 0.001$) with an average residual of -1.804×10^{-15} (range: -2.837 to 3.881, SD: 1.479) and an intercept of 3.594 ($p < 0.001$), as well as a negative relationship along the independence-submission axis ($\beta = -0.628$, $p < 0.001$) with an average residual of -8.177×10^{-16} (range: -2.491 to 3.670, SD: 1.492) and an intercept of 3.381 ($p < 0.001$). The observed co-concurrence of love and hate (or independence and submission), along with the presence of non-zero residuals—even if such residuals may partly result from measurement error or variability—raises the possibility that these dimensions may not be fully captured by a single-axis representation.

Additionally, we conducted a correlation analysis on participants' subjective ratings and found significant relationships between love and independence ($r = -0.512$, $p < 0.001$), hate and independence ($r = 0.557$, $p < 0.001$), and hate and submission ($r = -0.314$, $p = 0.012$). These results indicate that the original two axes, love–hate and independence–submission, may not be strictly orthogonal but rather interrelated.

To identify fundamental orthogonal dimensions of social recognition based on empirical data, we performed a principal component analysis (PCA) on the averaged ratings of the four categories across participants. Notably, we identified two pivotal components that collectively accounted for over 85% of the total variance observed in the behavioral ratings. The first component, PC1, accounting for 60.61% of the total variance, reflected a measure of social relationships, with love and submission positioned on one side of PC1, while hate and independence were positioned on the other (Fig. 1e). Consequently, PC1 could be interpreted as representing the Amity–Hostility dimension. The second component, PC2, explaining 24.43% of total variance, reflected a measure of more intricate social relationships, with love and independence on one end and hate and submission on the opposite end (Fig. 1e). This may denote a higher-level relationship that considers long-term benefits beyond the simplistic level indicated by PC1. Thus, PC2 can be considered to portray a Restrained Amity–Suppressive Hostility dimension. Additionally, we found PC3 and PC4, which explained 11.56% and 3.40% of the variance, respectively (Fig. 1e). We also assessed the agreement of the PCs across participants, and found that all four components (PC1, PC2, PC3 and PC4) were consistent ($t_{(29)} = 9.972$, $p < 0.001$ for PC1; $t_{(29)} = 8.196$, $p < 0.001$ for PC2; $t_{(29)} = 13.413$, $p < 0.001$ for PC3; $t_{(29)} = 8.494$, $p < 0.001$ for PC4) (Fig. 1f). Taken together, these results suggest the existence of two primary dimensions underlying social relationship recognition: the Amity–Hostility dimension and the Restrained Amity–Suppressive Hostility dimension, both of which were shared across participants.

Distinct processing of social recognition dimensions in the brain

We next investigated the neural bases underlying the processing of the two social recognition dimensions obtained from the behavioral ratings. For this, we first employed representational similarity analysis (RSA) combined with searchlight analysis[35–37]. Specifically, for each searchlight sphere, the representational dissimilarity matrix (RDM) was derived from the neural pattern similarity for each pair of clips and was directly correlated with the distance matrix between the clips along either the PC1 dimension (PC1-distance matrix) or the PC2 dimension (PC2-distance matrix) on the PC1-PC2 plane, which was derived from behavioral ratings (See "Methods" for details) (Fig. 2a).

Given that recognition of social relationships can occur both during and after viewing each clip, we applied this method to analyze neural responses elicited during the presentation of each clip (viewing phase), during the early phase of the subsequent delay period (early phase), immediately following clip presentation, and during the late phase of the delay period (late phase). Whole-brain RSA across the time phases revealed different temporal and spatial patterns between the processes of the social recognition dimensions. Specifically, regions encompassing the right temporal cortex and right parietal cortex, including the middle and posterior superior temporal sulcus (STS), the posterior cingulate cortex (PCC), and occipital cortical areas including the primary visual cortex (V1) were mainly involved in the processing of PC1 throughout the viewing phase, the early phase and the late phase (Fig. 2b, d and Supplementary Fig. 3a). On the other hand, strong processing of the PC2 was predominantly observed in the ventromedial prefrontal cortex (vmPFC), temporal cortical areas including the anterior STS (aSTS), and lateral occipital cortical areas during the late phase of the delay period, rather than earlier phases, although weaker PC2 processing was also detected in portions of medial cortical areas during the viewing phase (Fig. 2c, e and Supplementary Fig. 3b). The distribution of these observed neural substrates involved in PC1 and PC2 processing differed from those associated with the recognition of love, hate, independence, or submission, although some neural substrates overlapped (Supplementary Fig. 4). Collectively, these results suggest that the PC1 dimension undergoes rapid processing from the onset of clip viewing, with robust engagement of neural substrates such as the mSTS and pSTS throughout the delay period. In contrast, the processing of the PC2 dimension is characterized by dynamic changes in the involvement of neural substrates, with predominant engagement of areas such as vmPFC and aSTS occurring at a comparatively slower pace.

To conduct a more detailed statistical examination of the different processing of social recognition dimensions, we focused on specific regions, the rpSTS, the raSTS, and the rvmPFC, based on our whole-brain RSA results and prior findings[38–44]. This ROI-based RSA revealed the same tendency with the results from the whole-brain RSA. In rpSTS, we observed robust processing of the PC1 dimension throughout all phases ($t_{(29)} = 4.884$, $p < 0.001$ for viewing phase; $t_{(29)} = 2.589$, $p = 0.045$ for early phase; $t_{(29)} = 2.854$, $p = 0.036$ for late phase, FDR-corrected), whereas no significant processing of PC2 was detected (all $t_{(29)} < 1.224$, $p > 0.413$, FDR-corrected) (Fig. 3a, left). In rvmPFC, we only observed significant processing

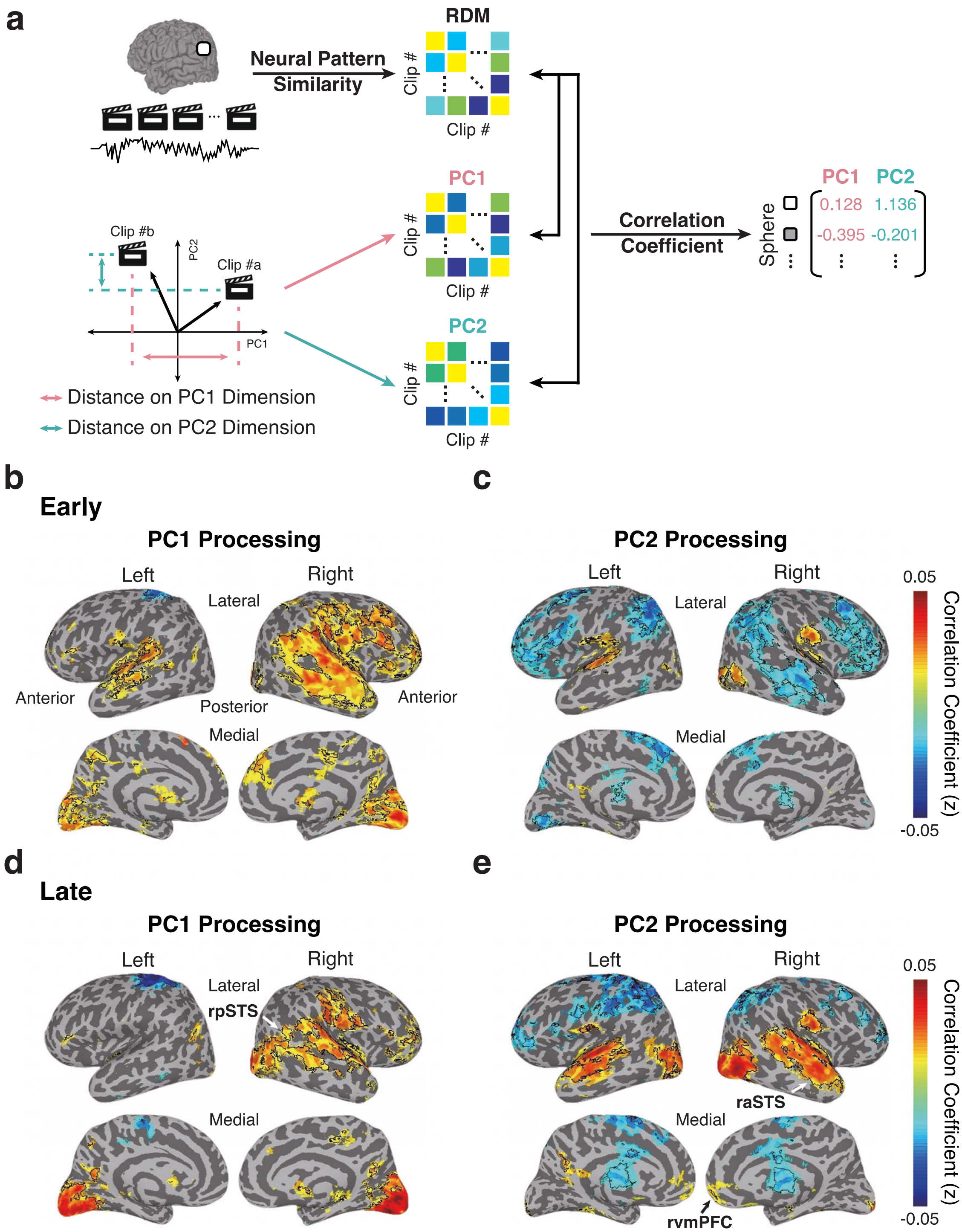

Fig. 2 | Neural substrates of dimensional processing in social recognition. **a** To identify the neural substrates involved in the dimensional processing of social recognition, we employed representational similarity analysis (RSA). Specifically, representational dissimilarity matrices (RDMs) derived from the neural response patterns for each pair of main clips were compared with distance matrices representing the relationship of clips along the PC1 dimension (PC1-distance matrix) or the PC2 dimension (PC2-distance matrix) on the PC1-PC2 plane, which was based on behavioral ratings. **b**, **c** Whole-brain maps show the RSA results for processing along the PC1 dimension (**b**) and PC2 dimension (**c**) during the early phase. Colored areas indicate significant clusters ($p < 0.05$, uncorrected), with black outlines indicating clusters that were corrected for multiple comparisons ($p < 0.05$, FDR-corrected). **d**, **e** Whole-brain maps show the RSA results for PC1 processing (**d**) or PC2 processing (**e**) during the late phase. Colored areas indicate significant clusters ($p < 0.05$, uncorrected), with black outlines indicating clusters that were corrected for multiple comparisons ($p < 0.05$, FDR-corrected).

of the PC2 dimension during the late phase ($t_{(29)} = 2.633$, $p = 0.044$, FDR-corrected), with no discernible processing of the PC1 dimension (all $t_{(29)} < 2.373$, $p > 0.063$, FDR-corrected) (Fig. 3a, right). In raSTS, significant processing of PC1 was observed during the viewing phase ($t_{(29)} = 2.925$, $p = 0.036$, FDR-corrected), while significant processing of PC2 was observed during the late phase ($t_{(29)} = 2.983$, $p = 0.036$, FDR-corrected) (Fig. 3a, middle). A two-way ANOVA with social recognition dimension (PC1, PC2) and temporal phase (viewing, early, late) as factors revealed a

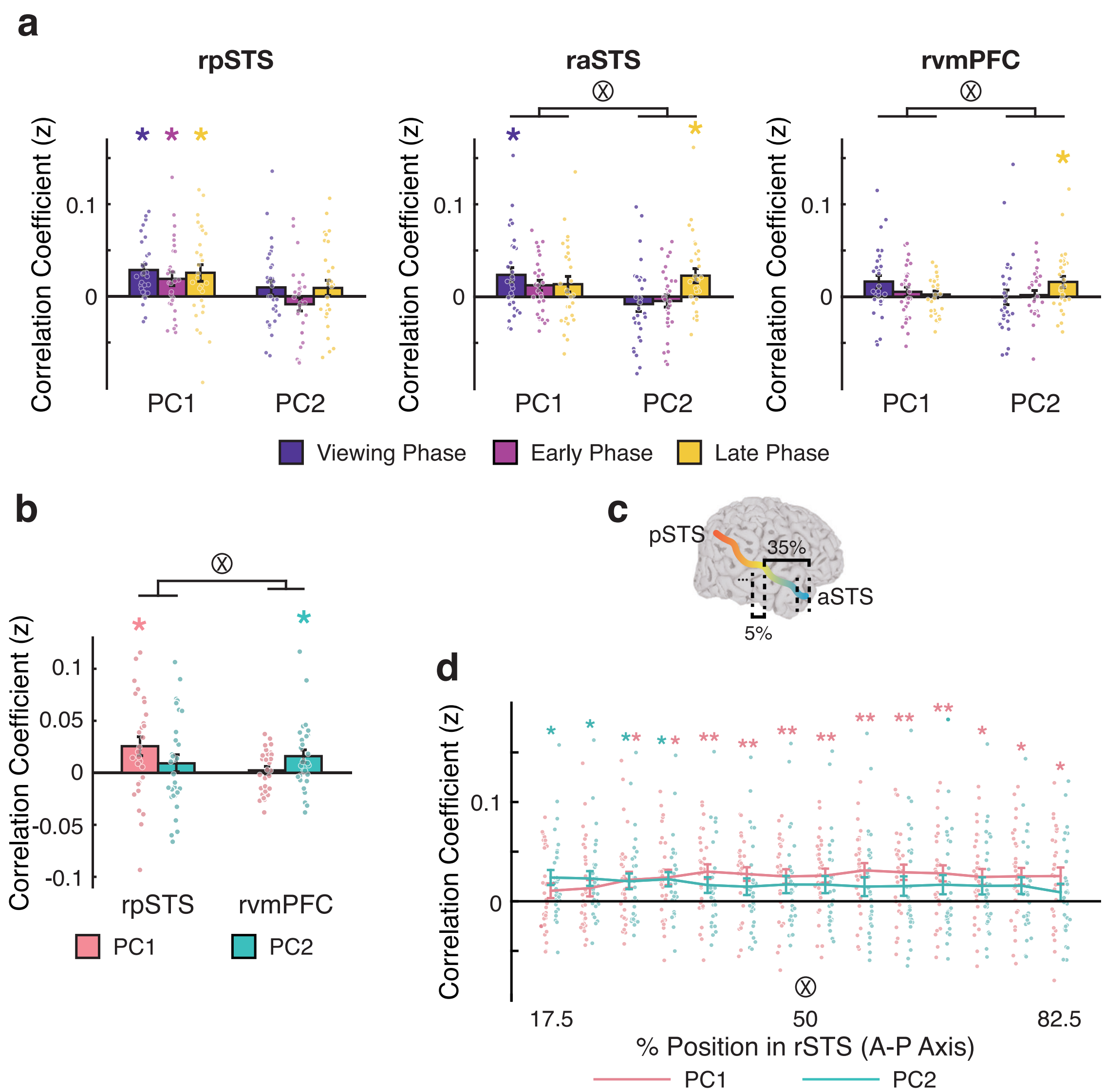

Fig. 3 | Distinct neural processing of social recognition dimensions. a Results from regions-of-interest (ROI)-based representational similarity analysis (RSA). The mean correlation coefficients (Fisher's z-transformed) for PC1 and PC2 processing during the viewing, early, and late phases are shown for the right posterior superior temporal sulcus (rpSTS), right anterior STS (raSTS), and right ventromedial prefrontal cortex (rvmPFC). Each dot represents the mean correlation for an individual participant. Error bars indicate the between-subject s.e.m. **b** The mean correlation coefficients (Fisher's z-transformed) for PC1 and PC2 processing during the late phase in the rpSTS and rvmPFC. The values in this graph correspond to those in (**a**) and were presented to facilitate comparison with subsequent graphs. **c** To examine the PC1 and PC2 processing along the anterior-posterior axis of the rSTS, we defined subregions, where each subregion accounted for 35% of the total length of the rSTS, shifting the center from the anterior to the posterior end. The subregions were centered at every 5% of the slices of the rSTS along the anterior-posterior axis. **d** The mean correlation coefficients (Fisher's z-transformed) for PC1 and PC2 processing during the late phase along the anterior-posterior axis of the rSTS. Each dot represents the mean correlation for an individual participant. Error bars depict the between-subject s.e.m. $^{**}p < 0.01$, $^{*}p < 0.05$ (t-test, two-tailed, FDR-corrected). A symbol ⊗ indicates a significant interaction effect (two-way ANOVA, $p < 0.05$).

significant interaction between dimension and phase in the raSTS ($F_{(1.473,\ 42.717)} = 6.316$, $p = 0.008$) and rvmPFC ($F_{(1.445,\ 41.917)} = 5.526$, $p = 0.014$), indicating different temporal dynamics for the PC1 and PC2 dimensions in these regions. We also examined the medial parietal cortical areas, the Precuneus (PCu) and the Posterior Cingulate Cortex (PCC), but no significant PC1 or PC2 processing was detected (Supplementary Fig. 5). Given the robust processing of both PC1 and PC2 dimensions during the late phase of the delay period, albeit with stronger PC1 processing during earlier phases, our subsequent analyses focused on the processing during the late phase.

Since distinct processing of PC1 and PC2 was more prominent in rpSTS and rvmPFC compared to other areas, we further examined the separate processing of PC1 and PC2 dimensional information focusing on these regions during the late phase. A two-way ANOVA with social recognition dimension (PC1, PC2) and ROI (rpSTS, rvmPFC) as factors revealed significant interaction between dimension and ROI ($F_{(1,\ 29)} = 5.933$, $p = 0.021$) (Fig. 3b). This result indicates distinct processing of PC1 and PC2 dimensions in rpSTS and rvmPFC.

We also replicated this result using multiple regression analysis. To do this, we regressed each neural RDM against the PC1-distance matrix and the PC2-distance matrix (See "Methods" for details) (Supplementary Fig. 6a). Consistent with the correlation result, we observed distinct processing of PC1 and PC2 in the rpSTS and the rvmPFC (Supplementary Fig. 6b). A two-way ANOVA revealed a significant interaction between dimension and ROI ($F_{(1,\ 29)} = 5.563$, $p = 0.025$). The significant processing of PC1 ($t_{(29)} = 2.815$, $p = 0.035$, FDR-corrected) not PC2 ($t_{(29)} = 1.020$, $p = 0.421$, FDR-corrected) was detected in rpSTS, whereas the opposite tendency was found in rvmPFC ($t_{(29)} = 2.526$, $p = 0.035$, for PC2; $t_{(29)} = 0.390$, $p = 0.700$, for PC1, FDR-corrected). We also examined the processing of PC1 and PC2 dimensions in the pSTS and the vmPFC of the left hemisphere, but did not observe any significant processing (Supplementary Fig. 7).

Additionally, since significant processing of PC1 (but not PC2) was observed in the rpSTS, whereas the opposite tendency was found in the raSTS during the late phase of the delay period (Fig. 3a), we examined PC1 and PC2 processing along the anterior-posterior axis of the rSTS (Fig. 3c). Consistent with the ROI-based RSA results, PC1 processing was predominantly observed in the middle and posterior regions of rSTS (all $t_{(29)} > 2.657$, $p < 0.024$ for the positions between the position of 32.5% of slices and the posterior end, FDR-corrected), while significant PC2 processing was more prominent in the anterior regions of the rSTS (all $t_{(29)} > 2.559$, $p < 0.028$ for the positions between the anterior end and the position of 37.5% of slices, FDR-corrected) (Fig. 3d). Multiple regression analysis corroborated these findings (Supplementary Fig. 6c). Furthermore, a two-way ANOVA with dimension (PC1, PC2) and position (first slice, last slice) as factors revealed a significant interaction between them in both analyses ($F_{(1, 29)} = 8.757$, $p = 0.006$ for correlation; $F_{(1, 29)} = 14.371$, $p < 0.001$ for multiple regression) (Fig. 3d and Supplementary Fig. 6c). These results suggest distinct processing of PC1 and PC2 along the anterior-posterior axis of the rSTS.

Our RSA results showed a significant distinction in the processing of PC1 and PC2 dimensions, particularly evident in the rpSTS and rvmPFC regions. We further validated these results using support vector regression (SVR) (Fig. 4a). Based on above RSA results, it is expected that PC1 dimension may be predominantly predicted from the neural response patterns of rpSTS, while PC2 dimension can be mainly predicted from the neural response patterns of rvmPFC. To test this, we first applied PCA to the neural response patterns across voxels in the rpSTS and rvmPFC to reduce the dimensionality of the neural responses, and then used the resulting principal components (PC1 to PC37) to predict the PC1- and PC2-values of social recognition across clips (Fig. 4a). The SVR performance was evaluated by comparing the predicted PC1- and PC2-values with the observed PC values (See "Methods" for details) (Fig. 4a). Consistent with our RSA results, significant PC1 processing ($t_{(29)} = 3.227$, $p = 0.006$, FDR-corrected), but not PC2 ($t_{(29)} = -0.107$, $p = 0.915$, FDR-corrected), was predicted from the neural response patterns of the rpSTS (Fig. 4b). Conversely, significant PC2 processing ($t_{(29)} = 3.337$, $p = 0.006$, FDR-corrected), but not PC1 ($t_{(29)} = 1.621$, $p = 0.155$, FDR-corrected), was predicted from the neural response patterns of the rvmPFC (Fig. 4b). Moreover, a significant interaction effect between dimension (PC1 and PC2) and ROI was observed (two-way ANOVA, $F_{(1, 29)} = 6.137$, $p = 0.019$), indicating distinct processing of PC1 and PC2 in the rpSTS and rvmPFC. The significance of these findings was further confirmed through permutation tests (Fig. 4c, d).

These results collectively show that the two fundamental dimensions of social recognition involve different temporal dynamics and spatial distinctions in neural processing. These suggest that while PC1 is processed relatively immediately and consistently in the cortical regions, particularly the right posterior superior temporal sulcus, PC2 involves more dynamic changes in neural substrates and primarily engages the right ventromedial prefrontal cortex at a slower pace.

Neural networks coupled with the distinct processing of social recognition dimensions

Having observed that the rpSTS and the rvmPFC were distinctively involved in social recognition dimensions, we next examined the distribution of the regions that communicate with either the rpSTS or the rvmPFC during social recognition. To do this, we explored which regions exhibit close representational relationships with the rpSTS or the rvmPFC, using representational connectivity analysis[36,45,46]. This approach allowed us to identify brain regions across the whole brain that are functionally connected to the rpSTS and rvmPFC and engaged in information processing similar to that of these regions. We directly compared the RDMs derived from each seed ROI (rpSTS or rvmPFC) with the RDMs derived from each searchlight voxel[35] (Fig. 5a). Notably, posterior cortical regions, including the middle and posterior temporal areas, parietal regions, and the occipital areas, exhibited significantly stronger correspondence with the rpSTS. In contrast, anterior cortical regions, including the lateral and medial prefrontal areas and anterior temporal areas, showed stronger correspondence with the rvmPFC (Fig. 5b and Supplementary Fig. 8). These results suggest that the areas primarily interacting with the rpSTS and those mainly interacting with the rvmPFC are differently distributed across the anterior and posterior regions of the cortex.

To further clarify the relative interaction with the rpSTS and the rvmPFC, we focused on the neural representations of V1 and vlPFC, which showed strong correspondence with the rpSTS and the rvmPFC, respectively, in the searchlight analysis (Fig. 5b). We examined the representational relationships between the rpSTS and V1, the rpSTS and vlPFC, the rvmPFC and V1, and the rvmPFC and vlPFC (Fig. 5c). Consistent with the searchlight analysis, we found that the representation of V1 corresponded more strongly with the rpSTS than with the rvmPFC ($t_{(29)} = 16.725$, $p < 0.001$), whereas the representation of the vlPFC was closer to the rvmPFC than to the rpSTS ($t_{(29)} = 2.623$, $p = 0.014$) (Fig. 5d). Moreover, a significant interaction effect between the seed ROIs and target ROIs was observed (two-way ANOVA, $F_{(1, 29)} = 91.296$, $p < 0.001$), indicating different strengths of interaction between them.

If there were a close representational relationship between the rpSTS and V1, and between the rvmPFC and vlPFC, then the neural representation of V1 may be predicted from that of the rpSTS, while the neural representation of the rvmPFC may be predicted from that of the vlPFC. We tested this idea using the SVR method (Fig. 5e) (See "Methods" for details). As expected, the observed neural representation of V1 was significantly closer to the predicted representation from the rpSTS than from the rvmPFC ($t_{(29)} = 5.397$, $p < 0.001$), while the observed representation of the vlPFC was closer to the predicted representation from the rvmPFC than from the rpSTS ($t_{(29)} = 4.773$, $p < 0.001$) (Fig. 5f). Moreover, a significant interaction effect was found between the seed ROIs and target ROIs (two-way ANOVA, $F_{(1, 29)} = 41.900$, $p < 0.001$) (Fig. 5f). Furthermore, we applied the same SVR procedure with the seed ROIs and target ROIs reversed, and observed the same interaction effect (two-way ANOVA, $F_{(1, 29)} = 43.895$, $p < 0.001$), with better prediction of rpSTS representation from V1 than from vlPFC ($t_{(29)} = 2.438$, $p = 0.021$), and better prediction of rvmPFC representation from vlPFC than from V1 ($t_{(29)} = 8.498$, $p < 0.001$) (Fig. 5g).

Collectively, our findings suggest that the two principal dimensions underlying social relationship recognition are predominantly processed in distinct neural networks: a posterior cortical network centered on the pSTS for PC1, and an anterior cortical network centered on the vmPFC for PC2.

Discussion

Our findings suggest that individuals recognize social relationships primarily based on two fundamental dimensions. The first dimension, reflecting the Amity–Hostility property, was predominantly processed in posterior cortical regions, particularly the right posterior STS (rpSTS), whereas the second dimension, reflecting the Restrained Amity–Suppressive Hostility property, was mainly processed in anterior cortical regions, including the right ventromedial prefrontal cortex (rvmPFC). Collectively, these results suggest that the recognition of complex human social relationships is based on distinct neural processing of these two primary dimensions in the brain.

Our results suggest that humans commonly recognize social relationships based on two fundamental dimensions. The first dimension (PC1) represents a measure of social relationships, with love and submission positioned on one end and hate and independence on the other (Fig. 1e), which can be interpreted as the Amity–Hostility dimension. Since submitting to a loving relationship and avoiding an adversarial one is likely to enhance individual interests and survival, it is reasonable that this dimension is processed with primary importance. The second dimension (PC2) reflects a more intricate assessment of social relationships, with love and independence on one end and hate and submission on the opposite (Fig. 1e),

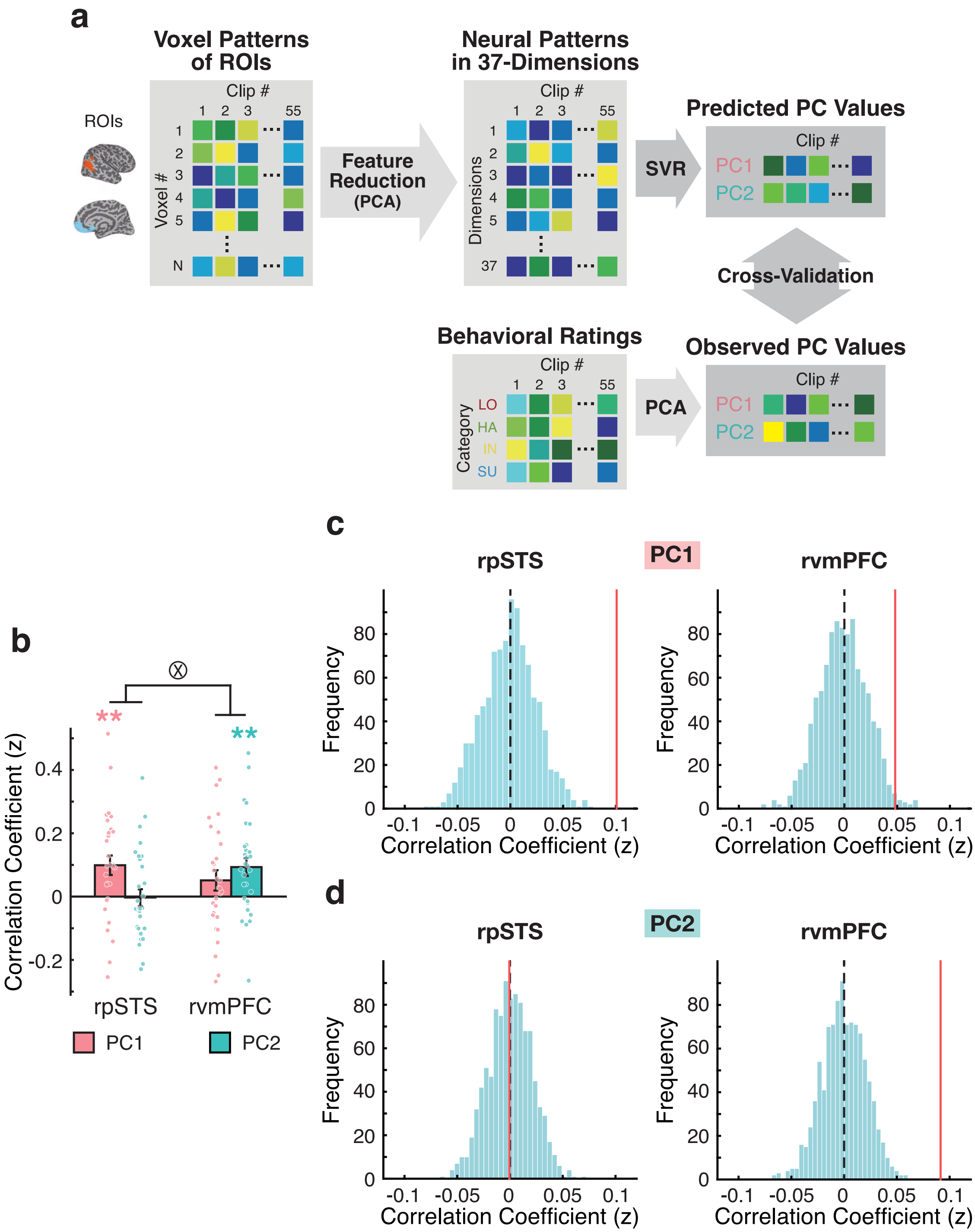

Fig. 4 | Decoding dimensional states of social recognition from neural response patterns of the rpSTS and the rvmPFC. a The potential to predict the PC1- and PC2-values of social recognition across clips from the neural response patterns of the rpSTS and rvmPFC was assessed using support vector regression (SVR). The dimensionality of the neural responses from each ROI was first reduced through principal component analysis (PCA). The resulting principal components (PC1 to PC37) were then utilized in SVR to predict the PC1- and PC2-values. SVR performance was evaluated by comparing the predicted PC1- and PC2-values with the observed values obtained from behavioral ratings. **b** The mean correlation between the predicted PC values from the SVR and the observed PC values were calculated. Each dot represents the average correlation value for each participant. $^{**}p < 0.01$ (t-test, two-tailed, FDR-corrected). A symbol ⊗ indicates a significant interaction effect (two-way ANOVA, $p < 0.05$). Error bars indicate between-subjects s.e.m. **c**, **d** Results of the permutation test for PC1 (**c**) and PC2 (**d**) based on SVR. Histograms illustrate the null distribution of correlation values generated from the SVR procedure described in (**a**), using randomly shuffled PC values across 1000 iterations. Red vertical lines indicate the actual average correlation coefficients across participants.

which can be interpreted as the Restrained Amity–Suppressive Hostility dimension. This dimension appears to relate to adhering to social norms and being considerate of others, potentially serving to maintain the stability of the social system rather than focusing solely on individual benefit.

In this study, participants were asked to simultaneously rate the degree of love, hate, submission, and independence for each given clip of a naturalistic movie. This design allowed us to empirically examine whether the axes of love versus hate and submission versus independence, proposed in

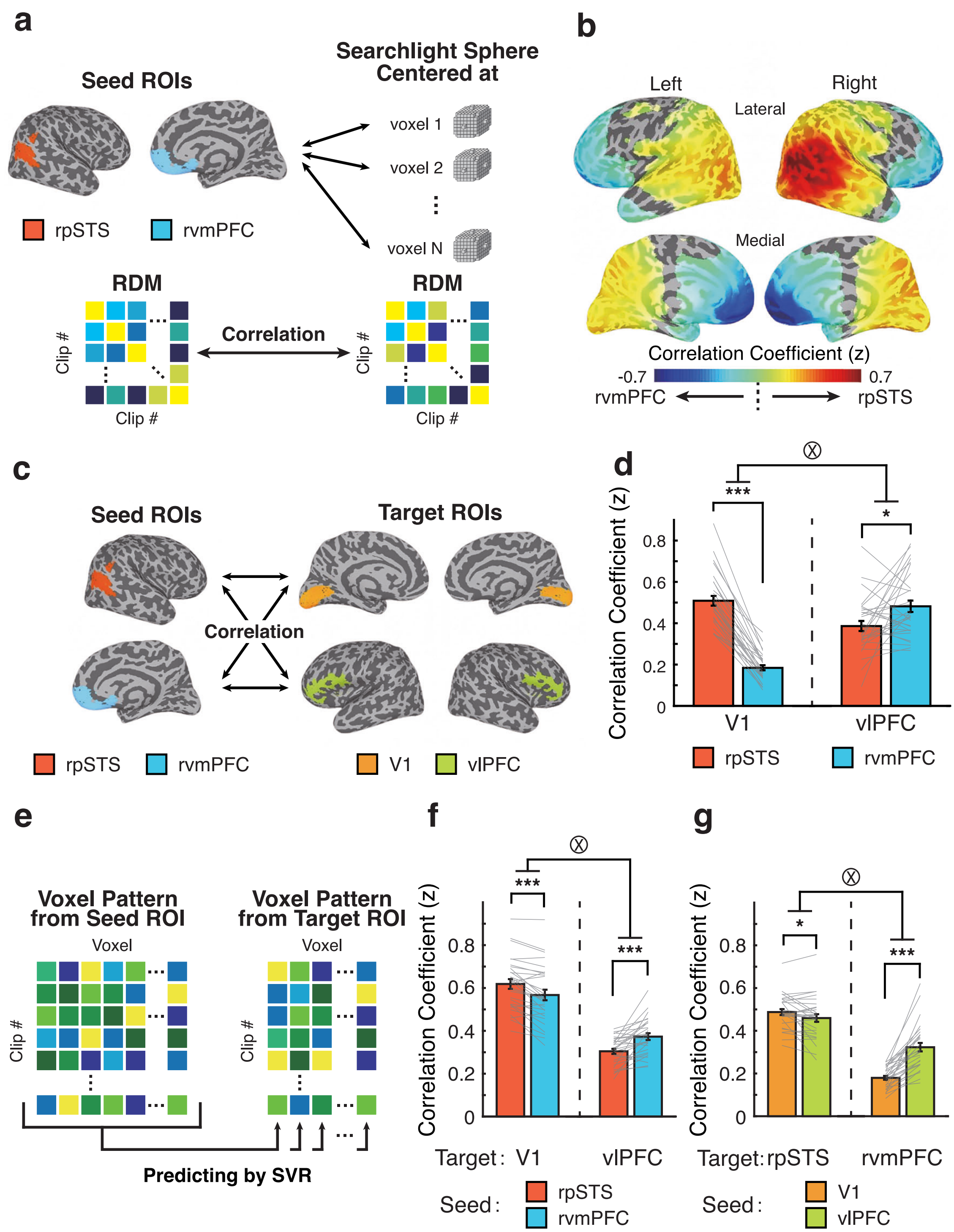

Fig. 5 | Representational connectivity with the rpSTS and the rvmPFC. **a** To investigate brain regions showing close representational relationships with the rpSTS or the rvmPFC, the RDM derived from each seed ROI (rpSTS or rvmPFC) was correlated with the RDM from each searchlight sphere. **b** A whole-brain map shows the difference in representational connectivity results for each seed ROI, the rpSTS and the rvmPFC ($p < 0.001$, FDR-corrected). **c** To assess the representational relationships between seed ROIs and target regions, the RDM derived from each seed ROI was correlated with the RDM from each target ROI. **d** The results of the representational connectivity analysis using rpSTS or rvmPFC as a seed for each target region, the primary visual cortex (V1) and the ventrolateral prefrontal cortex (vlPFC). The endpoints of each gray line indicate the correlation values for each participant. **e** The voxel-wise neural pattern of the target ROI was predicted using SVR based on the neural pattern derived from the seed ROI. **f** The performance of SVR in predicting the neural pattern of the target ROIs, the V1 and the vlPFC, from the neural patterns of the seed ROIs, the rpSTS and the rvmPFC. The endpoints of each gray line indicate SVR performance for each participant. **g** The performance of SVR for predicting the neural patterns of the rpSTS and rvmPFC using the neural patterns of the V1 and the vlPFC. The endpoints of each gray line indicate SVR performance for each participant. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$ (t-test, two-tailed, FDR-corrected). A symbol ⊗ indicates a significant interaction effect (two-way ANOVA, $p < 0.05$). Error bars indicate between-subjects s.e.m.

previous studies[1–3,6], would emerge as the principal dimensions along which people recognize social relationships. However, the two social recognition dimensions we identified differed from prior theoretical models. Love and hate, as well as submission and independence, were not processed as singular dimensions of affiliation or autonomy, nor were they orthogonal to each other. Instead, the two dimensions we found (PC1 and PC2) represented a more intertwined relationship between love, hate, independence, and submission. This finding aligns with recent reports[21,34] suggesting that the affiliation and the autonomy axes are interconnected. It is possible that rather than processing affiliation (love–hate) and autonomy (submission–independence) separately and then combining them, the brain may handle these dimensions in a more integrated form, like PC1 and PC2, from the outset. This approach may enable more efficient information processing.

Because our findings were based on a single movie, one might argue that PC1 and PC2 dimensions we identified may not generalize to other social stimuli. To address this concern, we conducted additional analyses. Specifically, we applied two approaches: (1) grouping clips featuring the same designated actor and then randomly dividing each group into two subsets, or (2) splitting all clips randomly into two subsets. PCA was then conducted separately on the behavioral ratings from each subset. In both cases, the PC dimensions remained highly consistent with those derived from the full set of movie clips (Supplementary Fig. 9). These results suggest that the identified dimensions are not driven solely by particular clips, but rather reflect a more general and stable structure underlying social relationship recognition. Nevertheless, further research using a broader range of stimuli and a more extensive set of rating dimensions of social relationships would be valuable to further support and extend our findings.

Our findings reveal distinct neural substrates underlying the processing of the two fundamental dimensions of social recognition. Posterior cortical areas, particularly rpSTS, were primarily involved in processing PC1, whereas anterior cortical areas, including rvmPFC, were mainly engaged in processing PC2 (Figs. 2d, e, 3a, b, 4b). Previous studies have implicated the STS and mPFC in perceiving social interactions between individuals and/or empathizing with others' thoughts (theory of mind)[39–42,47–53]. Additionally, recent electrocorticography (ECoG) research shows a stronger gradient of mentalizing processes from visual cortex to the mPFC, and suggests that while self- and other-mentalizing recruit similar regions, other-mentalizing elicits slower and more prolonged activations[54]. These prior findings are consistent with our results regarding the recruited ROIs in social relationship recognition and the delayed engagement of the mPFC. However, the distinct roles of the pSTS and the mPFC in social recognition have remained unclear. Our results provide evidence that these areas are differentially involved in processing distinct dimensions of social relationship recognition. Furthermore, given the distributed neural substrates underlying social recognition processing[16–18,21–23,25,55,56], our connectivity analyses reveal network-level contributions to the processing of social recognition dimensions beyond the two key regions. Specifically, anterior cortical regions primarily contributed to processing the PC2 dimension, while posterior cortical regions were more engaged in processing the PC1 dimension (Fig. 5b). Collectively, these findings suggest a neural framework for social recognition that operates through information-based processing along two primary dimensions.

The first principal dimension (PC1) we found was predominantly processed in posterior cortical regions, including visual cortex (V1) and rpSTS. These regions are known to be critical in perceiving social features, such as facial expressions, and are considered to be part of the third visual pathway, which is proposed to be specialized for social perception[27–29]. Furthermore, PC1 dimension was processed in these areas from the viewing phase to the late phase (Fig. 3a), indicating rapid and sustained processing. Based on these findings and previous studies [27–29], we propose that the first fundamental dimension, PC1, is associated with social feature perception, which is processed rapidly in response to social stimuli in the neural network encompassing the visual cortex and rpSTS.

Empathizing with the mental states or intentions of others through theory of mind processing, in conjunction with perceiving social features through the sensory system, is critical for recognizing social relationships. The vmPFC has been implicated in theory of mind processing[42,49,57], but it has also been reported to play a role in the perception of social interactions[52,58–60]. Given that the vmPFC is also involved in retrieving self-referential information, such as autobiographical memories or schemas[61–70], it is plausible that neural processing in this region supports the understanding of others' intentions or mental states based on information recalled from one's own experience. Additionally, the vmPFC, vlPFC, and their connectivity have been associated with emotional regulation[71–77] and social pain processing[78–83]. This regulation process may support social relationships by prioritizing long-term or group benefits over immediate individual preferences. Our results showed that the vmPFC is prominently involved in processing the PC2 dimension, particularly during the late phase rather than the earlier phases (Figs. 2e and 3a, b). Furthermore, there was significant representational connectivity between the vmPFC and vlPFC (Fig. 5d). These results, combined with previous research, suggest that PC2 dimensional information is mainly linked to theory of mind and emotional regulation. This type of processing may require more time than PC1, which is processed more immediately based on social feature perception. These findings introduce a concept of integrating social feature perception processing with theory of mind processing. Future research is needed to explore the mechanisms by which PC1 and PC2 dimensions are integrated to facilitate the recognition of social relationships.

In conclusion, our findings suggest that the recognition of complex human social relationships is processed based on two primary dimensions: the Amity–Hostility dimension and the Restrained Amity–Suppressive Hostility dimension, each engaging distinct neural systems in the brain.

Methods

Participants

Thirty neurologically intact right-handed native Korean participants (15 identified as female and 15 as male; age = 22.30 ± 2.02 years) took part in the fMRI experiment. One additional participant was excluded because the movie clips could not be presented in the correct story order. All participants provided a written informed consent for the procedure in accordance with protocols approved by the Seoul National University and KAIST Institutional Review Board.

Stimuli and task design

Participants performed a social recognition task inside an fMRI scanner. During the task, they watched movie clips (390 × 219 pixels, 12 × 6.75 °) edited from the film *Actress* (J-yong E, Showbox, 2009) and were asked to rate the interpersonal relationships between the actor and the target person depicted in each movie clip, from the actor's perspective (Fig. 1b).

Each trial began with a 6.5–108.0 second movie clip (henceforth referred to as the "context clip" or "context", duration = 35.78 ± 25.67 s), which provided background information about the social situation for the subsequent clip for rating. Following the context clip, a red fixation cross (onset cue) appeared along with a beep, signaling to the participants that the instruction phase would soon follow. The red cross was displayed for 2.07 ± 0.56 s, with a duration varying between 1.0 s and 3.0 s to synchronize the delay phase with the beginning of the TR. The beep sound lasted for 0.5 s.

In the subsequent instruction phase, participants were informed of the identities of the actor and the target by displaying their faces (6 × 6 °) on either side of the fixation cross (the actor was positioned 4.5 ° to the left of the center, and the target on the opposite side) for 3 s. This was followed by a 1 s inter-stimulus interval (ISI) during which a yellow fixation cross was presented. After the ISI, a movie clip for rating the social relationship recognition (henceforth referred to as the "main clip" or "clip", duration = 8.97 ± 4.76 s, ranging from 2.4 s to 23.2 s) was shown. Following the main clip, a 5 s-delay period ensued, followed by the rating phase. During the rating phase, participants evaluated the main clip based on four categories–love, hate, independence, and submission–using an eleven-point scale (0: Did not recognize this category, 5: Recognized this category, 10: Highly recognized this category) (Fig. 1b). They had a maximum of 12 s to complete the rating. Once a rating was made, the screen transitioned to the inter-trial interval (ITI). The ITI lasted a minimum of 2 s, and the total duration of the ITI combined with the rating time was fixed at 14 s.

A total of 63 main clips, each with a corresponding context clip, were presented during the task. Of these 55 main clips were included in the analyses because the targets of the excluded clips were unclear (the targets were not persons but specific situations). All clips were arranged sequentially, allowing for a natural narrative flow from one trial to the next. This design, which preserved situational and contextual continuity, was essential for the recognition of interpersonal relationships, particularly those between the designated Actor and Target. During the task, each clip was presented only once, without repetition, to minimize potential confounding influences from specific events or learning effects. The task consisted of six runs, each comprising 9 to 13 trials.

Behavioral data analyses

To assess the consistency in social relationship recognition across participants for the main clips, we calculated Spearman's correlation coefficients between subjective ratings for each of the four categories: love, hate, independence, and submission. The correlation coefficients were then transformed using Fisher's z-transformation and averaged across all possible participant pairs (Fig. 1c).

To identify the principal dimensions of social recognition, we performed a principal component analysis (PCA) on the averaged ratings of the four categories across participants (Fig. 1e). To evaluate the consistency of the resultant principal components (PCs) across participants, we implemented a leave-one-out cross-validation procedure (Fig. 1f), following the approach used in previous studies[84]. Specifically, for each iteration, PCA was conducted on the ratings from the left-out subject and on the averaged ratings from the remaining participants. The four components obtained from each left-out subject were rotated (reflection only) to match those derived from the rest of the participants. This procedure was repeated for each participant, and the component weights were compared across participants using Spearman's correlation. The resulting correlation coefficients were then transformed using Fisher's z-transformation.

Clustering analysis of the averaged ratings across participants was conducted in two phases. First, the behavioral ratings were reduced to two dimensions using the Barnes-Hut t-SNE algorithm (t-Distributed Stochastic Neighbor Embedding; perplexity = 30; theta = 0.5)[85,86]. Subsequently, the reduced ratings were clustered using the k-means clustering algorithm[87].

To assess whether the principal dimensions we identified were influenced by specific clips or reflected a more generalizable pattern, we split the clips into two separate groups and conducted PCA on each subset independently (Supplementary Fig. 9). We tested two grouping methods: (1) random assignment of clips into two subsets, and (2) grouping clips by the same designated actor (i.e., clips featuring the same actor were grouped together) followed by random assignment of these groups into two subsets. We then evaluated whether the resultant principal components were consistent between subsets and with those derived from the full set of movie clips. The four principal components obtained from each subset were rotated (reflection only) to match those derived from the original dimensions. Each grouping method was repeated for 5000 times, and the weights for the components were compared using Spearman's correlation.

fMRI data acquisition

Participants were scanned on the 3 T Siemens MAGNETOM Prisma located at the Center for Neuroscience Imaging Research of the Institute for Basic Science. Images were acquired using a 64-channel head coil with an in-plane resolution of 2.5 × 2.5 mm and forty-eight 2.5 mm slices (0.25 mm inter-slice gap, repetition time (TR) = 2000 ms, echo time (TE) = 17 ms, field of view (FOV) = 192 mm, and flip angle = 90 °). Standard MPRAGE (magnetization-prepared rapid-acquisition gradient echo) were collected with an in-plane resolution of 1 × 1 mm, and 1 mm slices (TR = 2300 ms, TE = 2.28 ms, FOV = 256 mm, flip angle = 8 °) at the end of the experiment runs for each participant.

fMRI data analysis

The data analysis was conducted using AFNI[88], SUMA (AFNI surface mapper)[89], FreeSurfer[90], and custom MATLAB scripts. Data preprocessing included slice-time correction and motion correction, after with the percent signal change was calculated for each scan run and each participant on a voxel-by-voxel basis.

To deconvolve event-related responses, we employed a standard general linear model using the AFNI software package ('3dDeconvole' with 'BLOCK' option). Given that each clip was presented without repetition across the trials of experiment, each trial in the experiment was modeled as a separate event. The t-values for each clip (trial) and voxel were derived for the viewing phase (when the clip was presented), the early phase of the subsequent delay period (early phase), and the late phase of the delay period (late phase). The early and late phases corresponded to the first and second halves of the 5-second delay period, respectively.

Representational similarity analysis (RSA)

To elucidate the neural substrates underlying the dimensional processing of social recognition, we employed representational similarity analysis (RSA)[36,37]. Specifically, during the viewing, early phase, and late phases, we calculated Pearson correlations between voxel patterns for each pair of clips using the estimated t-values for each clip across voxels. Representational dissimilarity matrices (RDMs) were then derived from the Fisher's z-transformed correlation coefficients between the clips.

The RSA was conducted by correlating the RDM derived from the neural response patterns with a matrix representing the relationships between clips (main clips) based on the PC1 or PC2 dimension. These clip relationships were calculated from the Euclidean distances between each pair of clips on the PC1-PC2 plane and were normalized by dividing the distances by the maximum value of each dimension (Fig. 2a). The PC1-distance matrix and the PC2-distance matrix, which reflect the distances between the clips along the PC1 and PC2 dimensions, respectively, were derived. Correlation coefficients were then calculated by comparing the PC1-distance matrices or the PC2-distance matrices to the RDMs derived from voxel response patterns in each searchlight sphere (radius = 11.25 mm, corresponding to ~341 voxels) across the brain (Fig. 2)[35] or in each region-of-interest (ROI) (Fig. 3). The resulting correlation coefficients were Fisher's z-transformed. This procedure was also applied to compare the RDMs with distance matrices from the clip relationships based on each of the four rating categories (love, hate, independence, and submission) (Supplementary Fig. 4).

We also replicated the correlation-based RSA results using a multiple regression approach[91]. Specifically, we regressed each neural RDM against the PC1-distance matrix and the PC2-distance matrix (Supplementary Fig. 6a). We used generalized linear regression implemented in MATLAB (using the fitglm function) to assess the simultaneous effects of both principal components on the neural patterns by regressing the neural RDM with the two distance matrices. As a result of the multiple linear regression, we obtained β coefficients for each component to evaluate their respective effects. To compare the coefficients for each PC in each ROI, we converted the β coefficients into t-scores by dividing them by their respective standard errors.

Support vector regression (SVR)

Using support vector regression (SVR), we assessed the ability to predict the PC1- and PC2-values of social recognition based on neural response patterns in each ROI (Fig. 4a). To do this, we first applied principal component analysis (PCA) to reduce the dimensionality of voxel-wise neural response patterns within each ROI for each participant. The first 37 principal components (PC1 ~ PC37), accounting for over 85% of the total variance, were then used as input features for the SVR model (MATLAB, fitrsvm function).

To predict values and evaluate SVR model performance, we implemented a 100×10-fold cross-validation procedure. In each iteration, the reduced neural response dimensions (PC1 ~ PC37) for the 55 main clips were randomly partitioned into 10 subsamples (5 ~ 6 clips per subsample) using MATLAB's cvpartition function. One subsample was designated as the test set, while the remaining nine subsamples were used for training. This procedure was repeated 10 times, with each subsample designated as the test set once, resulting in predictions of PC1- and PC2-values of social recognition for all 55 main clips. SVR performance was evaluated by comparing the predicted PC1- and PC2-values with the observed values from behavioral ratings. For this, Spearman's correlation coefficients were calculated between the predicted and observed PC1- or PC2-values for the 55 clips. These correlation values were then transformed into Fisher's z-scores. The prediction and evaluation procedure was repeated 100 times, and the resultant z-scores were averaged across iterations for every participant. The statistical significance of SVR performance was assessed by generating a null distribution of z-scores from 1000 repetitions of the prediction and evaluation procedure, using randomly shuffled PC1- and PC2-values of social recognition.

Representational connectivity analysis

To assess representational similarity between regions, we employed representational connectivity analysis[36,45] using two approaches. First, we applied a searchlight analysis[35], in which the RDM, reflecting dissimilarities between neural response patterns of the clips, was derived from a seed ROI and correlated with the RDM from each searchlight sphere across the brain (Fig. 5a). Second, we performed ROI-based comparisons, where the RDM from each seed ROI was correlated with the RDM from each target ROI (Fig. 5c).

To test whether the voxel-wise neural response patterns of the target ROI could be predicted from the seed ROI, or vice versa, we employed an SVR method using MATLAB's fitrsvm function (Fig. 5e). We predicted voxel-wise neural response patterns and evaluated SVR model performance using a leave-one-out cross-validation procedure, where voxel patterns from one clip was used as the test set, and patterns from the remaining 54 clips served as the training set. For each voxel in the target ROI, an SVR model was trained to predict the t-value of the voxel based on voxel patterns of the seed ROI in the training set. The trained SVR model was then applied to the test set to predict the t-values of voxels in the target ROI from the voxel patterns of the seed ROI. SVR performance was assessed by comparing the predicted and observed t-values of the target ROI, using Pearson's correlation coefficients, which were subsequently transformed into Fisher's z-scores. This procedure was repeated 55 times, with each clip serving as the test set once. The resulting z-scores were averaged across iterations for each participant.

Regions-of-interest (ROIs)

Regions-of-Interest (ROIs) were defined using the cortical parcellation from FreeSurfer[90]. The label "S_temporal_sup" was used to identify the superior temporal sulcus (STS)[92]. The posterior superior temporal sulcus (pSTS) was defined by the posterior 33% of slices along the anterior-posterior axis. For the ventromedial prefrontal cortex (vmPFC), the "medialorbitofrontal" label was used[93], and the primary visual cortex (V1) was defined using the "V1" label. The ventrolateral prefrontal cortex (vlPFC) was characterized by combining regions associated with the labels "G_front_inf-Opercular", "G_front_inf-Orbital", and "G_front_inf-Triangul"[92]. To examine the neural responses along the anterior-posterior axis of the right STS (rSTS), subregions were defined, each accounting for 35% of the total rSTS length, with centers shifted incrementally by 5% of the slices from anterior to posterior.

Statistics and reproducibility

To assess agreement between participants' ratings for each category or between participants' principal components, two-tailed paired t-tests were performed, comparing the results to a baseline of zero. To compare correlation coefficients in RSA, SVR, or representational connectivity analyses (RCA), two-tailed paired t-tests were conducted, with comparisons made against a baseline of zero. P values were adjusted for multiple comparisons across temporal phases, dimensions, and/or ROIs (or spheres in RCA) using false discovery rates (FDR) correction, based on the Benjamini-Hochberg procedure[94], implemented through MATLAB's mafdr function. Repeated-measures ANOVAs (within-subject effects) were used to determine statistical significance in comparisons of the average number of clips and average rating level in behavioral ratings, as well as ROI, dimension, and interaction effects. For all two-way ANOVAs, Sphericity Assumed correlations were used if the assumption of sphericity was met; Otherwise, Greenhouse-Geisser corrections were used.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The processed fMRI data are available under restricted access due to limitations in the informed consent obtained from participants, and the raw fMRI and MRI data are protected and are not available due to data privacy laws. The sharing and reuse of the data are available upon request from the corresponding author and require approval form the Institutional Review Boards. Source data are provided with this paper. The source data behind the graphs in the paper can be found in Supplementary Data.

Code availability

The custom MATLAB scripts used to analyze the functional MRI data in this study are available on GitHub at https://github.com/memcolab/Kang2025CommunBiol.

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References

1. Wiggins, J. S. A psychological taxonomy of trait-descriptive terms: The interpersonal domain. *J. Pers. Soc. Psychol.* **37**, 395–412 (1979).
2. Birtchnell, J. The Interpersonal Octagon: An Alternative to the Interpersonal Circle. *Hum. Relat.* **47**, 511–529 (1994).
3. Kiesler, D. J. Contemporary interpersonal theory and research: personality, psychopathology, and psychotherapy. *J. Psychother. Pract. Res.* **6**, 339 (1997).
4. Birtchnell, J. Relating therapy. *Br. J. Psychother.* **30**, 87–100 (2014).
5. Benjamin, L. S., Rothweiler, J. C. & Critchfield, K. L. The use of structural analysis of social behavior (SASB) as an assessment tool. *Ann. Rev. Clin. Psychol.* **2**, 83–109 (2006).
6. Benjamin, L. S. Structural analysis of social behavior. *Psychol. Rev.* **81**, 392–425 (1974).
7. Fiske, S. T., Cuddy, A. J. C. & Glick, P. Universal dimensions of social cognition: warmth and competence. *Trends Cognit. Sci.* **11**, 77–83 (2007).
8. Simon, J. C., Styczynski, N. & Gutsell, J. N. Social perceptions of warmth and competence influence behavioral intentions and neural processing. *Cogn. Affect Behav. Neurosci.* **20**, 265–275 (2020).
9. Premack, D. & Woodruff, G. Does the chimpanzee have a theory of mind? *Behav. Brain Sci.* **1**, 515–526 (1978).
10. Van Overwalle, F. & Baetens, K. Understanding others' actions and goals by mirror and mentalizing systems: A meta-analysis. *NeuroImage* **48**, 564–584 (2009).
11. Kennedy, D. P. & Adolphs, R. The social brain in psychiatric and neurological disorders. *Trends Cognit. Sci.* **16**, 559–572 (2012).
12. Rizzolatti, G. & Sinigaglia, C. The functional role of the parieto-frontal mirror circuit: Interpretations and misinterpretations. *Nat. Rev. Neurosci.* **11**, 264–274 (2010).

13. Keysers, C. & Gazzola, V. Integrating simulation and theory of mind: from self to social cognition. *Trends. Cognit. Sci.* **11**, 194–196 (2007).
14. Redcay, E. & Schilbach, L. Using second-person neuroscience to elucidate the mechanisms of social interaction. *Nat. Rev. Neurosci.* **20**, 495–505 (2019).
15. Gallese, V., Fadiga, L., Fogassi, L. & Rizzolatti, G. Action recognition in the premotor cortex. *Brain* **119**, 593–609 (1996).
16. Molenberghs, P., Cunnington, R. & Mattingley, J. B. Brain regions with mirror properties: A meta-analysis of 125 human fMRI studies. *Neurosci. Biobehav. Rev.* **36**, 341–349 (2012).
17. Rizzolatti, G. & Craighero, L. The mirror-neuron system. *Ann. Rev. Neurosci.* **27**, 169–192 (2004).
18. Caspers, S., Zilles, K., Laird, A. R. & Eickhoff, S. B. ALE meta-analysis of action observation and imitation in the human brain. *Neuroimage* **50**, 1148–1167 (2010).
19. Malle, B. F. Attribution theories: How people make sense of behavior. *Theories in Social Psychology*, Second Edition, 93–120 (2022).
20. Kelley, H. H. & Michela, J. L. Attribution theory and research. *Annu Rev. Psychol.* **31**, 457–501 (1980).
21. Tavares, R. M. et al. A map for social navigation in the human brain. *Neuron* **87**, 231–243 (2015).
22. Parkinson, C., Liu, S. & Wheatley, T. A common cortical metric for spatial, temporal, and social distance. *J. Neurosci.* **34**, 1979–1987 (2014).
23. Muscatell, K. A. et al. Social status modulates neural activity in the mentalizing network. *Neuroimage* **60**, 1771–1777 (2012).
24. Thornton, M. A., Weaverdyck, M. E., Mildner, J. N. & Tamir, D. I. People represent their own mental states more distinctly than those of others. *Nat. Commun.* **10**, 2117 (2019).
25. Buckner, R. L. & Carroll, D. C. Self-projection and the brain. *Trends Cogn. Sci.* **11**, 49–57 (2007).
26. Saxe, R., Carey, S. & Kanwisher, N. Understanding other minds: Linking developmental psychology and functional neuroimaging. *Annu Rev. Psychol.* **55**, 87–124 (2004).
27. Pitcher, D. & Ungerleider, L. G. Evidence for a third visual pathway specialized for social perception. *Trends Cognit. Sci.* **25**, 100–110 (2021).
28. McMahon, E. & Isik, L. Seeing social interactions. *Trends Cognit. Sci.* **27**, 1165–1179 (2023).
29. McMahon, E., Bonner, M. F. & Isik, L. Hierarchical organization of social action features along the lateral visual pathway. *Curr. Biol.* **33**, 5035–5047.e8 (2023).
30. Sliwinska, M. W. & Pitcher, D. TMS demonstrates that both right and left superior temporal sulci are important for facial expression recognition. *Neuroimage* **183**, 394–400 (2018).
31. Pitcher, D., Sliwinska, M. W. & Kaiser, D. TMS disruption of the lateral prefrontal cortex increases neural activity in the default mode network when naming facial expressions. *Soc. Cogn. Affect. Neurosci*. **18**, (2023).
32. Wiggins, J. S., Phillips, N. & Trapnell, P. Circular reasoning about interpersonal behavior: evidence concerning some untested assumptions underlying diagnostic classification. *J. Personal. Soc. Psychol.* **56**, 296 (1989).
33. Fiske, S. T. Warmth and competence: Stereotype content issues for clinicians and researchers. *Can. Psychol.* **53**, 14–20 (2012).
34. Todorov, A., Said, C. P., Engell, A. D. & Oosterhof, N. N. Understanding evaluation of faces on social dimensions. *Trends Cogn. Sci.* **12**, 455–460 (2008).
35. Kriegeskorte, N., Goebel, R. & Bandettini, P. Information-based functional brain mapping. *Proc. Natl. Acad. Sci. USA* **103**, 3863–3868 (2006).
36. Kriegeskorte, N., Mur, M. & Bandettini, P. Representational similarity analysis - connecting the branches of systems neuroscience. *Front Syst. Neurosci.* **2**, 249 (2008).
37. Haxby, J. V. et al. Distributed and overlapping representations of faces and objects in ventral temporal cortex. *Science (1979)* **293**, 2425–2430 (2001).
38. Mar, R. A. The neural bases of social cognition and story comprehension. *Annu Rev. Psychol.* **62**, 103–134 (2011).
39. Deen, B., Koldewyn, K., Kanwisher, N. & Saxe, R. Functional organization of social perception and cognition in the superior temporal sulcus. *Cereb. Cortex* **25**, 4596–4609 (2015).
40. Masson, H. L., & Isik, L. Functional selectivity for social interaction perception in the human superior temporal sulcus during natural viewing. *Neuroimage* **245**, 118741 (2021).
41. Isik, L., Koldewyn, K., Beeler, D. & Kanwisher, N. Perceiving social interactions in the posterior superior temporal sulcus. *Proc. Natl. Acad. Sci. USA* **114**, 9145–9152 (2017).
42. Bzdok, D. et al. Segregation of the human medial prefrontal cortex in social cognition. *Front Hum. Neurosci.* **7**, 232 (2013).
43. Hiser, J. & Koenigs, M. The multifaceted role of the ventromedial prefrontal cortex in emotion, decision making, social cognition, and psychopathology. *Biol. Psychiatry* **83**, 638–647 (2018).
44. Varrier, R. S. & Finn, E. S. Seeing social: A neural signature for conscious perception of social interactions. *J. Neurosci.* **42**, 9211–9226 (2022).
45. Freund, M. C., Etzel, J. A. & Braver, T. S. Neural coding of cognitive control: the representational similarity analysis approach. *Trends Cognit. Sci.* **25**, 622–638 (2021).
46. Pillet, I., Op de Beeck, H. & Lee Masson, H. A comparison of functional networks derived from representational similarity, functional connectivity, and univariate analyses. *Front Neurosci.* **13**, 1348 (2020).
47. Beauchamp, M. S. The social mysteries of the superior temporal sulcus. *Trends Cognit. Sci.* **19**, 489–490 (2015).
48. Lahnakoski, J. M. et al. Naturalistic fMRI mapping reveals superior temporal sulcus as the hub for the distributed brain network for social perception. *Front. Hum Neurosci* https://doi.org/10.3389/fnhum.2012.00233 (2012).
49. Decety, J. & Sommerville, J. A. Shared representations between self and other: A social cognitive neuroscience view. *Trends Cogn. Sci.* **7**, 527–533 (2003).
50. Krueger, F., Barbey, A. K. & Grafman, J. The medial prefrontal cortex mediates social event knowledge. *Trends Cogn. Sci.* **13**, 103–109 (2009).
51. Yang, D. Y. J., Rosenblau, G., Keifer, C. & Pelphrey, K. A. An integrative neural model of social perception, action observation, and theory of mind. *Neurosci. Biobehav. Rev.* **51**, 263–275 (2015).
52. Pine, D. S., Wise, S. P. & Murray, E. A. Evolution, emotion, and episodic engagement. *Am. J. Psychiatry* **178**, 701–714 (2021).
53. Chin, J. H., Haring, K. S. & Kim, P. Understanding the neural mechanisms of empathy toward robots to shape future applications. *Front Neurorobot* **17**, 1145989 (2023).
54. Tan, K. M. et al. Electrocorticographic evidence of a common neurocognitive sequence for mentalizing about the self and others. *Nat. Commun.* **13**, 1919 (2022).
55. Thornton, M. A., Weaverdyck, M. E. & Tamir, D. I. The brain represents people as the mental states they habitually experience. *Nat. Commun.* **10**, 2291 (2019).
56. Saxe, R. & Kanwisher, N. People thinking about thinking people: The role of the temporo-parietal junction in 'theory of mind'. *Neuroimage* **19**, 1835–1842 (2003).
57. Ferstl, E. C. & von Cramon, D. Y. What does the frontomedian cortex contribute to language processing: Coherence or theory of mind?. *Neuroimage* **17**, 1599–1612 (2002).
58. Schilbach, L. et al. Being with virtual others: Neural correlates of social interaction. *Neuropsychologia* **44**, 718–730 (2006).
59. Amodio, D. M. & Frith, C. D. Meeting of minds: The medial frontal cortex and social cognition. *Nat. Rev. Neurosci.* **7**, 268–277 (2006).
60. Iacoboni, M. et al. Watching social interactions produces dorsomedial prefrontal and medial parietal BOLD fMRI signal increases compared to a resting baseline. *Neuroimage* **21**, 1167–1173 (2004).

61. Macrae, C. N., Moran, J. M., Heatherton, T. F., Banfield, J. F. & Kelley, W. M. Medial prefrontal activity predicts memory for self. *Cereb. Cortex* **14**, 647–654 (2004).
62. Moore, W. E., Merchant, J. S., Kahn, L. E. & Pfeifer, J. H. 'Like me?': Ventromedial prefrontal cortex is sensitive to both personal relevance and self-similarity during social comparisons. *Soc. Cogn. Affect Neurosci.* **9**, 421–426 (2014).
63. Cabeza, R., St Jacques, P. Functional neuroimaging of autobiographical memory. *Trends Cogn. Sci*. **11**, 219–227 (2007).
64. Svoboda, E., McKinnon, M. C. & Levine, B. The functional neuroanatomy of autobiographical memory: A meta-analysis. *Neuropsychologia* **44**, 2189–2208 (2006).
65. Hassabis, D. & Maguire, E. A. The construction system of the brain. *Philos. Trans. R. Soc. B: Biol. Sci.* **364**, 1263–1271 (2009).
66. Addis, D. R., Wong, A. T. & Schacter, D. L. Remembering the past and imagining the future: Common and distinct neural substrates during event construction and elaboration. *Neuropsychologia* **45**, 1363–1377 (2007).
67. McCormick, C., Barry, D. N., Jafarian, A., Barnes, G. R. & Maguire, E. A. vmPFC drives hippocampal processing during autobiographical memory recall regardless of remoteness. *Cereb. Cortex* **30**, 5972–5987 (2020).
68. Bonnici, H. M. & Maguire, E. A. Two years later – Revisiting autobiographical memory representations in vmPFC and hippocampus. *Neuropsychologia* **110**, 159–169 (2018).
69. Hebscher, M. & Gilboa, A. A boost of confidence: The role of the ventromedial prefrontal cortex in memory, decision-making, and schemas. *Neuropsychologia* **90**, 46–58 (2016).
70. Gilboa, A. & Marlatte, H. Neurobiology of Schemas and Schema-Mediated Memory. *Trends Cognit. Sci.* **21**, 618–631 (2017).
71. Kohn, N. et al. Neural network of cognitive emotion regulation - an ALE meta-analysis and MACM analysis. *Neuroimage* **87**, 345–355 (2014).
72. Koban, L., Kross, E., Woo, C. W., Ruzic, L. & Wager, T. D. Frontal-brainstem pathways mediating placebo effects on social rejection. *J. Neurosci.* **37**, 3621–3631 (2017).
73. Morawetz, C., Bode, S., Derntl, B. & Heekeren, H. R. The effect of strategies, goals and stimulus material on the neural mechanisms of emotion regulation: A meta-analysis of fMRI studies. *Neurosci. Biobehav. Rev.* **72**, 111–128 (2017).
74. Vijayakumar, N., Cheng, T. W. & Pfeifer, J. H. Neural correlates of social exclusion across ages: A coordinate-based meta-analysis of functional MRI studies. *NeuroImage* **153**, 359–368 (2017).
75. Wang, H., Braun, C. & Enck, P. How the brain reacts to social stress (exclusion) – A scoping review. *Neurosci. Biobehav. Rev.* **80**, 80–88 (2017).
76. He, Z. et al. Critical role of the right VLPFC in emotional regulation of social exclusion: a tDCS study. *Soc. Cogn. Affect Neurosci.* **13**, 357–366 (2018).
77. Zhao, J. et al. The VLPFC versus the DLPFC in downregulating social pain using reappraisal and distraction strategies. *J. Neurosci.* **41**, 1331–1339 (2021).
78. Johnstone, T., Van Reekum, C. M., Urry, H. L., Kalin, N. H. & Davidson, R. J. Failure to regulate: counterproductive recruitment of top-down prefrontal-subcortical circuitry in major depression. *J. Neurosci.* **27**, 8877–8884 (2007).
79. Delgado, M. R., Nearing, K. I., LeDoux, J. E. & Phelps, E. A. Neural circuitry underlying the regulation of conditioned fear and its relation to extinction. *Neuron* **59**, 829–838 (2008).
80. Schiller, D. & Delgado, M. R. Overlapping neural systems mediating extinction, reversal and regulation of fear. *Trends Cognit. Sci.* **14**, 268–276 (2010).
81. He, Z. et al. The VLPFC-engaged voluntary emotion regulation: Combined TMS-fMRI evidence for the neural circuit of cognitive reappraisal. *J. Neurosci.* **43**, 6046–6060 (2023).
82. He, Z. et al. The right VLPFC and downregulation of social pain: A TMS study. *Hum. Brain Mapp.* **41**, 1362–1371 (2020).
83. Steward, T. et al. Dynamic neural interactions supporting the cognitive reappraisal of emotion. *Cereb. Cortex* **31**, 961–973 (2021).
84. Lettieri, G. et al. Emotionotopy in the human right temporo-parietal cortex. *Nat. Commun.* **10**, 5568 (2019).
85. Berrios, R., Totterdell, P. & Kellett, S. Eliciting mixed emotions: A meta-analysis comparing models, types and measures. *Front Psychol.* **6**, 428 (2015).
86. Van Der Maaten, L. & Hinton, G. Visualizing data using t-SNE. *J. Mach. Learn. Res.* **9**, (2008).
87. Kaufman, L. & Rousseeuw, P. J. Finding groups in data: An introduction to cluster analysis (Wiley Series in Probability and Statistics). *Eepe.Ethz.Ch* vol. 66 (1990).
88. Cox, R. W. AFNI: Software for analysis and visualization of functional magnetic resonance neuroimages. *Comput. Biomed. Res.* **29**, 162–173 (1996).
89. Saad, Z. S. & Reynolds, R. C. SUMA. *NeuroImage* **62**, 768–773 (2012).
90. Fischl, B. FreeSurfer. *NeuroImage* **62**, 774–781 (2012).
91. Diedrichsen, J. & Kriegeskorte, N. Representational models: A common framework for understanding encoding, pattern-component, and representational-similarity analysis. *PLoS Comput Biol.* **13**, 1005508 (2017).
92. Destrieux, C., Fischl, B., Dale, A. & Halgren, E. Automatic parcellation of human cortical gyri and sulci using standard anatomical nomenclature. *Neuroimage* **53**, 1–15 (2010).
93. Desikan, R. S. et al. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage* **31**, 968–980 (2006).
94. Benjamini, Y. & Hochberg, Y. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **57**, 289–300 (1995).

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Author contributions

W.K., M.K., and S.-H.L. designed the experiment. W.K. and M.K. performed the experiments. W.K. analyzed the data and prepared the figures. W.K. and S.-H.L. interpreted the data and wrote the manuscript. S.-H.L. supervised the research.

Competing interests

The authors declare no competing interests.

Additional information

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